

JAVASCRIPT LOOPS ASSIGNMENT

1. Print a Message 5 Times

Write a JavaScript program to print the following output 5 times using a for loop:

I am Batman
I am Batman
I am Batman
I am Batman
I am Batman

2. Print a Message with Counter

Write a JavaScript program to print the following output using a loop:

I am Batman 1
I am Batman 2
I am Batman 3
I am Batman 4
I am Batman 5
I am Batman 6
I am Batman 7
I am Batman 8
I am Batman 9

3. Print Numbers from 10 to 1

Write a JavaScript program to print numbers from 10 to 1 using:

- for loop
- while loop
- do...while loop

Expected Output:

10
9
8
7
6

5
4
3
2
1

4. Print "Hello World" 10 Times

Write a JavaScript program to print:

Hello World

10 times using a while loop.

5. Print Multiples of 5

Write a JavaScript program to print all the multiples of 5 between 1 and 100.

Implement the solution using:

- for loop
- while loop
- do...while loop

Expected Output:

5
10
15
20
25
...
100

6. Print Odd and Even Numbers

Write a JavaScript program to print all odd and even numbers between 1 and 100.

Expected Output:

Even Numbers:

2
4
6
...
100

Odd Numbers:

1
3
5
...
99

7. What Will Be the Output?

What will happen when the following JavaScript program is executed?

```
let i = 1;

while (i <= 1) {
  console.log("Hi Java");
}
```

Answer the following:

1. What will be the output?
2. How many times will "Hi Java" be printed?
3. Will the program terminate?
4. If not, why?

8. Print Characters with Character Codes

Write a JavaScript program to print:

- A-Z
- a-z
- 0-9

along with their respective character codes.

Example:

A = 65

B = 66

C = 67

...

Z = 90

and:

a = 97

b = 98

c = 99

...

z = 122

Use:

- for loop
- while loop

Hint:

You can use JavaScript methods such as:

`charCodeAt()`

`String.fromCharCode()`

9. Print the Following Series

Write JavaScript programs to print the following series.

A.

1.0

2.0

3.0

4.0

5.0

6.0

7.0

8.0

9.0

10.0

B.

0
9
18
27
36
45
54
63
72
81
90
99

Use a loop to generate the series instead of hardcoding the values.

10. Print Only Vowels

Write a JavaScript program to print only the vowels:

a
e
i
o
u

Start the loop from 'a' and continue until 'z'.

Use:

- for loop
- while loop

Hint:

`"aeiou".includes(character)`

11. Use break in a Loop

Write a JavaScript program to print numbers from 1 to 10.

However, when you reach the number whose multiplication by 7 is being evaluated, stop the loop and print:

Bye, see you tomorrow

The solution should demonstrate the use of:

`break;`

12. Print Numbers from 1 to 100

Write a JavaScript program to print numbers from 1 to 100 using a for loop.

13. Print Numbers Divisible by 3

Write a JavaScript program to print all numbers between 1 and 100 that are divisible by 3.

Expected Output:

```
3
6
9
12
...
99
```

14. Print Numbers Divisible by Both 3 and 5

Write a JavaScript program to print all numbers between 1 and 100 that are divisible by both 3 and 5.

Expected Output:

```
15
30
45
60
75
90
```

15. Calculate the Sum of Numbers

Write a JavaScript program to calculate the sum of numbers from 1 to 100.

Expected Output:

Sum = 5050

16. Calculate the Factorial of a Number

Write a JavaScript program to calculate the factorial of a given number using a loop.

Test Data:

Input: 5

Expected Output:

Factorial = 120

17. Reverse a Number

Write a JavaScript program to reverse a given number using a loop.

Test Data:

Input: 12345

Expected Output:

54321

18. Count the Number of Digits

Write a JavaScript program to count the number of digits in a given number.

Test Data:

Input: 123456

Expected Output:

Number of digits = 6

19. Print a Multiplication Table

Write a JavaScript program to print the multiplication table of a given number.

Test Data:

Input: 7

Expected Output:

$$7 \times 1 = 7$$

$$7 \times 2 = 14$$

$$7 \times 3 = 21$$

$$7 \times 4 = 28$$

$$7 \times 5 = 35$$

$$7 \times 6 = 42$$

$$7 \times 7 = 49$$

$$7 \times 8 = 56$$

$$7 \times 9 = 63$$

$$7 \times 10 = 70$$